Dear Quillstone Payments Hiring Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. I hold an MSc in Computer Science and work primarily in Python. I bring more than five years of experience across machine learning, NLP and research, including deploying models into production, and I am particularly interested in applying this experience to systems that help support teams respond to customers efficiently.

In my current role as a Machine Learning Engineer at Slavonica Document AI, I develop NLP systems for Croatian administrative documents. I built and deployed a document classifier that handles approximately 15,000 documents per day, and fine-tuned transformer models for classification and named entity recognition. I also implemented model serving with FastAPI and Docker on Azure, used Azure ML and MLflow for experimentation, model registry and MLOps workflows, and established production monitoring that helped detect and resolve an OCR change associated with a nine-point accuracy drop. This work has given me experience carrying models from data preparation and experimentation through serving and ongoing operation.

Earlier, as a Data Scientist at Baranjska kreditna banka, I built a customer-complaint topic-classification workflow, progressing from a TF-IDF and logistic-regression baseline to fastText. The resulting 11-topic classification supported weekly reporting to customer-service management. I also built a personal retrieval-augmented question-answering prototype over Croatian laws using sentence-transformers, FAISS and an open-weight LLM accessed through an API; it was a manually checked weekend prototype rather than a production system. I am also pursuing a PhD in computer science focused on named entity recognition in Croatian legal texts. Together, these experiences have shaped my practical approach to language-focused ML, from preparing text data to building and monitoring useful systems.

Quillstone's focus on AI for customer operations is a strong match for my background in NLP, customer-support text classification and production ML systems. I am drawn to the team's ownership, pragmatic approach and emphasis on learning from mistakes—principles that fit well with the careful monitoring and iterative improvement required for production ML. I would welcome the opportunity to discuss how my experience could contribute to your AI team.

Kind regards,
Petra Zlatarić-Novosel